Of the Standards of Science

Computational Modelling as Psychology's Common Language

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Abstract

The replication crisis in psychology has been attributed to incentive structures and statistical norms, but the role of researcher skill distribution has not been formally modelled. This thesis presents an agent-based computational model of a stylised scientific community in which fifty researchers, distributed across ten laboratories and ten research domains, produce and evaluate scientific models over 300 discrete time steps under evolutionary selection. Two scenarios are compared: a Specialist community, in which researchers have one pronounced domain expertise and weak secondary skills, and a Generalist community, in which researchers maintain a meaningful baseline across all domains alongside their primary expertise. Across 30 independent simulation runs per scenario, the specialist community produced a significantly higher replication failure rate (.58 vs. .55; Mann-Whitney $U = 688.0$, $p = .0004$, $d = 1.00$) and greater concentration of publications in fewer domains (Gini .130 vs. .083; $p < .0001$, $d = 1.70$). Researcher reputation did not differ significantly between scenarios ($p = .455$), but its composition differed structurally: specialists achieved equivalent career outcomes through fewer, higher-yield publications in their peak domain, while generalists produced more publications at lower per-event return. A pre-registered exploratory hypothesis that specialists would be more vulnerable to reputational debunking was reversed: generalist publications, distributed across domains at intermediate quality, were more susceptible to successful challenge than specialist publications concentrated in high-quality, high-expertise niches. Together, the results suggest that skill distribution is an independent structural determinant of replication rates, operating through mechanisms distinct from incentive reform, and that the communicative infrastructure of a discipline — how broadly researchers can evaluate work outside their primary expertise — shapes epistemic outcomes even when individual incentives are held constant.

Keywords: replication crisis, agent-based modelling, skill specialisation, publication bias, scientific communities

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Introduction

This thesis investigates a structural explanation for the replication crisis in psychology: the degree of skill specialisation among researchers. While prior work has examined how statistical norms (Smaldino & McElreath, 2016), incentive structures (Nissen et al., 2016), and methodological diversity (Devezer et al., 2019; Weisberg & Muldoon, 2009) shape scientific communities, the role of domain-level skill distribution — how narrowly or broadly researchers are trained across research areas — has not been formally modelled. To address this gap, an agent-based computational model is developed in which a stylised scientific community of fifty researchers, organised across ten laboratories and ten research domains, produces and evaluates scientific models over time. Two contrasting scenarios are compared: a *Specialist* community, in which researchers have a single pronounced peak skill and weak secondary competencies, and a *Generalist* community, in which researchers maintain a meaningful baseline across multiple domains alongside their primary expertise. All other structural parameters are held constant.

Four outcomes are examined across 30 independent simulation runs per scenario. Two outcomes — replication failure rate and domain concentration of the publication landscape — were pre-specified from theory before simulations were run and constitute the confirmatory core of the analysis. Two further outcomes — researcher reputation accumulation and vulnerability to debunking — were identified during model development and are treated as exploratory. The model does not aim to reproduce any specific empirical dataset; rather, it tests whether the structural contrast between specialist and generalist skill distributions is sufficient to generate differences in systemic outcomes consistent with known features of the replication crisis (Ioannidis, 2005; Open Science Collaboration, 2015). Results are assessed using non-parametric tests with effect size estimates, and the robustness of findings is examined through a one-at-a-time sensitivity analysis.

The Replication Crisis and Its Roots

Psychology is in a period of acute self-examination. The most visible symptom of this is the replication crisis: a large-scale coordinated effort by the Open Science Collaboration found that fewer than half of a representative sample of published psychological findings replicated when tested independently (Open Science Collaboration, 2015). This is not a problem unique to psychology — Ioannidis (2005) demonstrated formally that when base rates of true hypotheses are low and statistical power is modest, the majority of nominally significant published findings are expected to be false — but it has been felt with particular force in a discipline that derives much of its applied authority from the reliability of its empirical claims.

Two classes of explanation dominate the discussion. The first concerns incentive structures: academic careers are advanced by publications, and publications are advanced by significant results, which creates systematic pressure toward the kind of flexible analysis, selective reporting, and underpowered studies that inflate false-positive rates (Nissen et al., 2016; Smaldino & McElreath, 2016). The second, and arguably more fundamental, concerns the structure of psychological theorising itself. As Meehl (1978) argued decades before the replication crisis became a household term, the slow progress of soft psychology is rooted in the use of verbal theories that are flexible enough to accommodate almost any pattern of data after the fact. When a theory can be rendered consistent with contradictory findings by adjusting its informal interpretation, it generates no genuine predictions and cannot be falsified. It accumulates asterisks rather than knowledge.

These two problems are interrelated in a way that is easy to overlook. Verbal theories are not merely imprecise — they are structurally invisible to researchers in other labs. When the mechanism linking two variables is described only in natural language, two researchers who believe they are studying the same phenomenon may operationalise it so differently that replication is structurally impossible even before any incentive problem arises. A shared formal language would not eliminate researcher degrees of freedom, but it would make theoretical commitments explicit enough to be auditable, comparable, and cumulatively buildable. As Smaldino (2017) argues, computational models force the scientist to commit: assumptions are specified, parameters are quantified, and predictions follow mechanically rather than rhetorically. The very rigidity that makes a computational model seem limiting is what gives it communicative power.

The case for formal models, however, goes beyond the individual researcher. McElreath & Smaldino (2015) demonstrate through a population-level mathematical model that the same statistical parameters — low base rate, low power, high false-positive rate — that make individual findings unreliable also make individual replications unreliable. Under these conditions, not one but multiple successful replications are required before a result carries strong evidential weight, because replication can only function as an epistemological filter if the underlying detection process is reliable. Crucially, the dynamics of scientific communication — which findings get published, which replications are attempted, and how results diffuse through a field — determine whether the community converges on truth regardless of individual good intentions.

Devezer et al. (2019) extend this insight in a directly relevant direction. Using an agent-based model of a scientific community, they show that reproducibility and truth convergence are not equivalent properties, and that the *structure* of the research population matters as much as individual researcher behaviour. Populations dominated by conservative researchers who test only incremental variants of the current consensus achieve high reproducibility while spending little time at the true model. Populations with greater methodological diversity — including researchers willing to explore distant regions of model space — discover the true model faster and are more robust to getting stuck in local optima. Importantly, no single research strategy is uniformly optimal: epistemic diversity, rather than any particular method, is what buffers a scientific community against its own structural limitations (Weisberg & Muldoon, 2009).

The Theory Crisis: Why Formalization Alone Is Insufficient

The discussion so far might suggest that the remedy for psychology's problems is straightforward: replace verbal theories with formal ones. Eronen & Bringmann (2021) complicate this picture in an important way. They argue that psychology faces a theory crisis distinct from and deeper than the replication crisis: even in a hypothetical world where every finding replicated perfectly, psychological theories would still fail to cumulate, because the phenomena they describe are not stable enough to anchor formal models.

Three structural problems compound each other. First, psychological interventions are *fat-handed*: a manipulation designed to induce, say, cognitive load simultaneously affects arousal, motivation, working memory, and mood, making it impossible to isolate the mechanism of interest. Second, construct validity is chronically weak — the same label (e.g., "working memory", "self-esteem") refers to operationally distinct things across labs, so replication is structurally ambiguous even when it succeeds. Third, and most fundamentally, psychology lacks the bedrock of robust, theory-independent phenomena from which formal theories could be bottom-up constructed. Natural sciences progress in part because anomalies are sharp — a two-degree discrepancy in a measured constant is a crisis. In psychology, the effect sizes themselves are contested.

This is a genuine challenge for any argument in favour of computational modelling, including the one made in this thesis. The position taken here is nevertheless that Eronen & Bringmann (2021)'s critique sharpens rather than defeats the case for formal models. A computational model cannot conjure stable phenomena from an unstable empirical record, but it can make the *assumptions* about stability explicit — and explicit assumptions can be tested, refined, and challenged across labs in a way that implicit verbal assumptions cannot. This thesis does not argue that psychology can or should become a natural science in the classical sense; it argues that, whatever the ultimate stability of its phenomena, the communicative infrastructure of a discipline determines whether it converges on its best available models, and that computational formalisation improves that infrastructure.

Computational Models as a Common Language

Taken together, the findings reviewed above suggest that the replication crisis in psychology is not primarily a story of individual bad actors but of a field whose theoretical and communicative infrastructure is too informal to support cumulative progress. The appropriate response is not only procedural reform — pre-registration, power analysis, open data — but a deeper shift toward shared formal frameworks that make theoretical commitments visible, comparable, and testable across labs. Computational modelling is the most mature candidate for such a framework in psychology. It provides the quantitative precision that verbal theories lack, the mechanistic explicitness that enables genuine replication, and — as the work reviewed here demonstrates — the capacity to study the dynamics of scientific communities themselves as objects of inquiry.

Research Questions and Model Overview

The simulation model consists of three entity types — Labs, Researchers, and Scientific Models — interacting across ten research domains. Figure 1 provides a schematic overview of the model architecture and the causal relationships between its components.

Four research questions motivated the study design. Two were pre-specified from theory before simulations were run (H1, H2) and constitute the confirmatory core of the analysis. Two emerged during model development and are treated as exploratory (H3, H4).

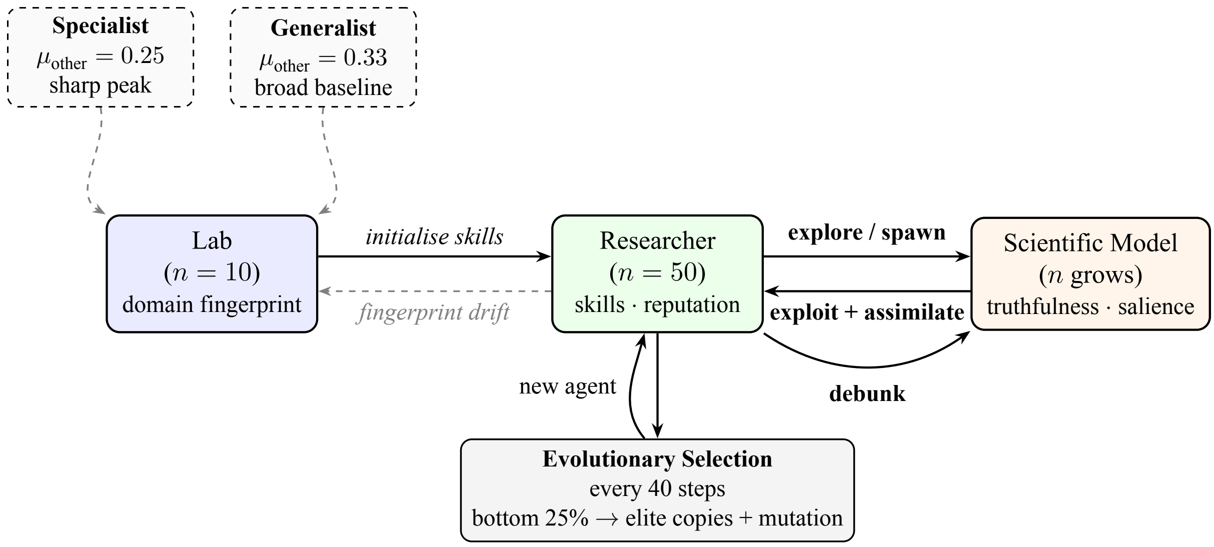

Figure 1

Schematic architecture of the agent-based model. Solid arrows indicate active causal processes; dashed arrows indicate passive or initialisation relationships. The two scenario boxes (top) define the only parameter that differs between conditions: μ_{other}, the mean non-peak skill of lab fingerprints. All other structural parameters are identical across both scenarios.

H1 (Replication failure rate). Does a specialist research community produce a higher rate of replication failure than a generalist community? Specialist researchers have a narrow skill profile, meaning their ability to successfully engage with models outside their primary domain is low. They are therefore more likely to fail replication attempts of models in areas where they lack competence, and less able to identify or correct errors in other domains. It is hypothesised that the specialist scenario will show a systematically higher replication failure rate than the generalist scenario.

H2 (Domain concentration). Does a specialist research community produce greater concentration of publications in fewer research domains? Because specialist researchers preferentially exploit models in their peak domain and have little incentive or ability to work across domains, the publication landscape is expected to become concentrated in a small subset of domains. Generalist researchers, by contrast, can profitably engage with a broader range of domains, producing a more evenly distributed literature. It is hypothesised that the specialist scenario will show a higher Gini coefficient of domain-level publication counts than the generalist scenario.

H3 (Researcher reputation — exploratory). Does community structure affect the average career success of individual researchers, as measured by accumulated reputation? Generalist researchers can accumulate publications across multiple domains and are less exposed to the saturation of any single domain, potentially enabling more consistent reputation growth. This hypothesis is treated as exploratory; the direction is plausible but not strongly constrained by theory.

H4 (Debunking stability — exploratory). Does community structure affect the degree to which researchers lose reputation through debunking events? Specialist researchers concentrate both their expertise and their publications in a single domain; successful debunking of a key model in that domain could therefore inflict proportionally larger reputational losses than the same event would in a generalist community. This hypothesis is treated as exploratory; the direction follows from the logic of concentration but the mechanism involves several countervailing forces that the model will adjudicate.

Methods

Simulation Environment

The simulation was implemented in Python using Mesa (Kazil et al., 2020), an open-source agent-based modelling framework. Mesa provides a standardised architecture for defining agents and model environments, a built-in data collector for recording agent- and model-level variables at each time step, and deterministic random number generation via a seeded NumPy generator, enabling exact reproduction of any run. The complete source code is publicly available at [REPOSITORY URL — to be inserted before submission].

Each simulation run proceeds for 300 discrete time steps. At each step, agents act in a randomly shuffled order, environmental dynamics are updated, and all outcome variables are recorded. Two scenarios are run consecutively under identical structural parameters, differing only in the distribution of initial researcher skill profiles (see [Scenarios]).

Model Architecture

The model represents a stylised scientific community composed of three types of entities: *Labs*, *Researchers*, and *Scientific Models*. Labs are static institutional objects. Researchers are active agents who learn, publish, and are subject to evolutionary selection. Scientific Models are passive environmental objects that accumulate citations and decay over time.

Labs

Ten laboratories define the institutional landscape of the simulation. Each lab is characterised by a fixed *domain fingerprint* — a vector of ten values specifying the lab's characteristic skill level in each of the ten research domains. The fingerprint determines the initial skill profiles of researchers hired into the lab and serves as an institutional anchor toward which researchers' skills drift over time (see [Core Mechanics]). Lab fingerprints are not updated through learning; institutional character is maintained by hiring.

At initialisation, each lab is assigned a randomly chosen *peak domain* in which its fingerprint value is drawn from a higher distribution, and nine *secondary domains* drawn from a lower distribution. The specific distribution parameters differ between the specialist and generalist scenarios (see Scenarios: Specialist vs. Generalist).

Researchers

Fifty researchers are distributed evenly across the ten labs (five per lab). Each researcher is characterised by a personal *domain skill vector* of ten values, one per research domain, each bounded in the interval [0.01, 0.95]. Initial skills are drawn from a normal distribution centred on the researcher's home lab's fingerprint, with a small noise term ($\sigma = 0.03$), so that researchers begin with profiles close to but not identical to their institutional template.

Researchers track publications, cumulative reputation, a count of steps spent in each activity mode, and their current research domain. Reputation is the primary currency of evolutionary selection: it accumulates through successful publications and decays passively at a rate of 0.2% per step, so that recent productivity counts more than historical output.

Scientific Models

A *Scientific Model* is a passive object representing a published scientific contribution within one of the ten domains. It carries four key attributes. Its *domain* identifies which research area it belongs to. Its *complexity vector* mirrors the domain fingerprint structure — a ten-element profile of how demanding the model is to work with across domains — drawn from the publishing researcher's current skill profile with noise ($\sigma = 0.05$). Its *truthfulness* ($\in [0, 1]$) measures scientific accuracy and productivity, as described below. Its *salience* ($\in [0, 1]$) measures current community attention — how visible and actively used the model is.

Truthfulness and domain caps. Each domain has a *truthfulness cap* drawn at initialisation from a uniform distribution over [0.35, 0.60], representing a ceiling on how accurate the best available model in that domain can currently be. When a new model is published, its truthfulness does not simply equal the researcher's skill; instead, it asymptotically approaches the domain cap. Specifically, a new model's truthfulness closes a fraction of the remaining gap between the current domain maximum and the cap, where the fraction is drawn from a Beta distribution whose shape parameter increases as the domain maximum grows. This produces diminishing returns: early models in an immature domain improve quickly; later models improve only incrementally. Caps grow continuously each step via asymptotic drift toward a hard ceiling of 0.95 (at rate 0.005 per step), so domains retain marginal scientific value throughout the simulation without abrupt discontinuities.

Salience dynamics. A model's salience begins at 0.5 at publication and is governed by two opposing forces. Each time another researcher successfully uses the model, it gains 0.05 salience (a citation). Each time step, salience decays at a base rate of 0.025. Crucially, high-truthfulness models resist decay as long as they are actively cited: the effective decay rate is reduced in proportion to the model's truthfulness, weighted by a *recency factor* that equals one when the model was cited in the current step and falls linearly to zero after 20 steps without a citation. A model that stops being cited therefore loses its truthfulness shield within 20 steps and then decays at the full base rate, regardless of how accurate it is. This creates era dynamics: superseded models fade on a predictable timescale and are gradually replaced by newer work. In addition, when a domain reaches high saturation ($\geq 70\%$ of its cap), all models in that domain incur an extra salience penalty each step, discouraging further investment in crowded areas.

Core Mechanics: Exploit, Invest, Explore, and Debunk

At each step, a researcher either continues an ongoing multi-step exploitation task or selects a new action. The four possible activities are described below.

Exploit. A researcher commits to implementing an existing Scientific Model. Implementation requires multiple consecutive steps, reflecting that engaging seriously with a published model takes sustained effort. The number of steps required scales with the model's complexity in its primary domain and the domain's current saturation, and is reduced by the researcher's own skill in that domain. When the required steps have been completed, the researcher makes an implementation attempt.

The probability of a successful attempt combines two factors: the researcher's raw proficiency in the model's domain, and a *similarity* term that measures how well the researcher's full skill profile matches the model's complexity profile, quantified as the Pearson correlation between the two vectors, floored at zero and weighted by domain proficiency. Formally, the similarity of researcher i to model m is

$$\text{sim}_{i,m} = \text{proficiency}_i(m) \times \text{max}\big(0,\ \text{Cor}(\mathbf{s}_i,\ \mathbf{c}_m)\big)$$

where $\mathbf{s}_i$ is the researcher's skill vector and $\mathbf{c}_m$ is the model's complexity vector. The overall success probability is then

$$P(\text{success}) = \Big(\text{sim}_{i,m} + \bar{s}_i\,(1 - \text{sim}_{i,m})\Big) \times \frac{\theta_{\text{actual}}}{\theta_{\text{reported}}} \tag{1}$$

where $\bar{s}_i$ is the researcher's mean skill across all domains and $\theta_{\text{actual}}/\theta_{\text{reported}}$ is the truth ratio. This ratio is the key mechanism through which publication bias enters replication dynamics: a model reported 20% above its actual quality is approximately 20% less likely to replicate successfully, regardless of researcher skill. A researcher who is skilled in the right domain and whose broader expertise aligns with the model's demands is most likely to succeed. The mean-skill floor ensures that any researcher with genuine competence always has a non-negligible success probability. The implementation in `researcher.py` reads directly:

```
def success_probability(self, m):
    sim = self._similarity(m)          # proficiency × max(0,
Cor(skills, complexity))
```

```
    avg = float(np.mean(self.domain_skills))
    base = sim + avg * (1.0 - sim)
    truth_ratio = m.actual_truthfulness / max(m.reported_truthfulness,
1e-8)
    return float(np.clip(base * truth_ratio, 0.0, 1.0))
```

On success, the researcher earns one publication, gains reputation proportional to the model's current *reported* truthfulness and salience (agents are rewarded based on what the community perceives, not the hidden actual quality), cites the model (raising its salience), and assimilates toward the model's complexity profile (see below). With a small probability that decreases as the domain fills up, the researcher also spawns a follow-up model, representing the opportunistic extension of successful work.

On failure, two additional processes are triggered based on the researcher's domain proficiency. First, an *expert truth correction* mechanism operates when proficiency exceeds 0.50: an expert who fails to replicate a result has enough domain knowledge to attribute the failure to the model's actual limitations rather than their own incompetence. Each such failure erodes the model's actual truthfulness by a small amount,

$$\Delta\theta_{\text{actual}} = -\,\text{proficiency} \times 0.015 \times \big(1 + \underbrace{(\theta_{\text{reported}} - \theta_{\text{actual}})}_{\text{bias gap}}\big) \qquad (2)$$

so that models with inflated reported quality erode faster. This creates a continuous, low-amplitude truth-revelation process distinct from the large discrete correction of a formal debunk. The implementation (`researcher.py`, failure branch of `_attempt`):

```
proficiency = self.domain_skills[target.domain]
if proficiency > 0.50:
    gap_inflation = target.reported_truthfulness -
target.actual_truthfulness
    correction    = proficiency * 0.015 * (1.0 + gap_inflation)
    target.actual_truthfulness = max(0.01, target.actual_truthfulness -
correction)
```

Second, a researcher with domain proficiency exceeding 0.35 has a probability proportional to their proficiency of initiating a formal debunk — the intuition being that an expert who cannot reproduce a result is well-positioned to mount a systematic challenge.

Invest. When a model's complexity in its primary domain exceeds the researcher's skill by more than a fixed training threshold, the researcher cannot yet attempt exploitation and instead invests in training. One training step moves the researcher's skills toward the model's complexity profile via the assimilation mechanism (see below), at a rate scaled by how well the researcher's overall profile already aligns with the model's demands.

Explore. The researcher publishes a new Scientific Model in a domain they select themselves. The value of exploring domain d is

$$V(\text{explore},\ d) = \frac{\text{skill}_d \times \max(0,\ 1 - \text{saturation}_d)}{1 + \ln(1 + n_d) + \text{own\_pubs}_d} \tag{3}$$

where n_d is the current number of models in domain d and own_pubs_d is the researcher's own prior publication count there. Saturated or personally exhausted domains offer little marginal value; fresh or underdeveloped domains are attractive targets.

A key extension to the basic explore action is the **breakthrough mechanic**. With a small probability that scales with the researcher's squared domain skill, the published model constitutes a paradigm-shifting breakthrough:

$$P(\text{breakthrough}) = \text{skill}_d^2 \times 0.10 \tag{4}$$

At skill 0.55 this yields approximately a 3% breakthrough rate; at skill 0.90 it is approximately 8%. A breakthrough model receives higher initial truthfulness (drawn from a Beta distribution concentrated near the top of the remaining gap-to-cap), starts with a salience of 0.90 rather than the standard 0.50, and triggers a *salience shock*: all incumbent models in the domain lose 65% of their current salience as the community's attention pivots to the new result. This creates era dynamics — trend cycles where a dominant paradigm is periodically displaced rather than accumulating indefinitely. The implementation (`researcher.py`, `_explore`):

```
skill = self.domain_skills[domain]
is_breakthrough = self.random.random() < skill ** 2 * 0.10

self.model.spawn_model(..., breakthrough=is_breakthrough)

if is_breakthrough:                             # salience shock to
incumbents
    new_uid = self.model.scientific_models[-1].uid
    for m in self.model.scientific_models:
        if m.domain == domain and m.uid != new_uid:
            m.salience = max(0.0, m.salience * 0.35)
```

Debunk. A researcher attempts to disprove an existing model. Debunking is not a freely chosen action; it is triggered as a byproduct of failed exploitation by a researcher with domain proficiency above 0.35. The probability of a successful debunk is

$$P(\text{debunk success}) = \text{proficiency} \times (1 - \theta_{\text{actual}}) \tag{5}$$

Experts are therefore most likely to expose weak models, and genuinely high-quality models are largely immune even when a very skilled researcher fails to replicate them. A successful debunk reduces the target model's truthfulness by 25% and its salience by 0.15, cascades partial damage to directly derivative models ($\theta_{\text{actual}} \times 0.88$ for children), and earns the researcher a publication and reputation proportional to the impact of the discredited model. The original author incurs a reputational penalty equal to half of the debunker's reward. The implementation (`researcher.py`, `_debunk`):

```
proficiency  = self.domain_skills[target.domain]
success_prob = proficiency * (1.0 - target.actual_truthfulness)
if self.random.random() < success_prob:
    target.actual_truthfulness = max(0.01, target.actual_truthfulness *
0.75)
    target.salience            = max(0.0,  target.salience - 0.15)
    # cascade to derivative models
    for m in self.model.scientific_models:
        if m.parent_uid == target.uid:
            m.actual_truthfulness = max(0.01, m.actual_truthfulness *
0.88)
```

Action selection. At the start of each step (if the researcher is not mid-exploitation), available models are partitioned into *exploit candidates* (complexity gap ≤ training threshold) and *invest candidates* (gap > threshold). The top two models by expected value are retained, and an explore option is added. The researcher then selects probabilistically among these options. The selection weight for each candidate is scaled by a *skill bias* term,

$$\text{skill\_bias}_d = \left(\frac{\text{skill}_d}{\bar{s}}\right)^{0.5} \tag{6}$$

where $\bar{s}$ is the researcher's mean skill. This is a parameter-free mechanism: a specialist with a skill ratio of ≈ 2 in their peak domain receives a weight multiplier of ≈ 1.41 for work in that domain, while a generalist with near-uniform skills receives a multiplier close to 1.0 everywhere. The bias emerges naturally from the skill distribution rather than being hardcoded, so it strengthens as specialisation deepens through assimilation. This ensures that both the intrinsic quality of a model and its current community visibility influence the researcher's choice, without making behaviour deterministic.

Assimilation. Working on a model gradually shifts the researcher's skills toward the model's complexity profile. The skill update follows an S-curve:

$$\Delta\text{skill}_d = r \times \text{skill}_d \times (1 - \text{skill}_d) \times 4 \times \max(0,\ c_d - \text{skill}_d) \times f_d \tag{7}$$

where r is the base learning rate (0.06 for attempts, 0.08 for training), c_d is the model's complexity in domain d, and f_d is a focus weight ($f_d = 1$ for the model's primary domain, $f_d = 0.12$ for all other domains). The gain is largest when current skill is near 0.5 and approaches zero at the extremes (no foundation to build on, or diminishing returns near mastery).

Learning is focused on the model's primary domain; secondary domains receive only 12% of the primary learning rate, preventing general engagement from silently inflating every skill. Skills only move upward through assimilation — working on a difficult model cannot reduce existing competence.

Fingerprint drift. To represent the pull of institutional norms, each researcher's non-peak skills drift passively toward their home lab's fingerprint every step. Skills above the fingerprint baseline decay at 0.4% per step; skills below it are pulled upward at 0.2% per step. The researcher's peak domain is exempt, allowing genuine specialisation to grow freely.

Evolutionary Selection and Lab Turnover

Every 40 steps, an evolutionary selection event occurs. Researchers are ranked by the reputation they have accumulated since the previous selection event — recent performance, not career totals. The bottom 25% are removed and replaced by mutated copies of elite researchers (the top 25% by the same criterion). Mutation perturbs each skill in the child's profile by a small amount ($\sigma = 0.04$), introducing variation while preserving the general shape of high-performing profiles. Each replacement researcher joins the same lab slot as the removed researcher, so the population size and lab distribution remain constant.

Following each selection event, the model evaluates whether any lab's average recent performance falls below a threshold relative to the field average. If the weakest lab's mean per-researcher reputation gain is less than 50% of the overall mean, that lab's fingerprint is replaced with a newly drawn random fingerprint. Current researchers in the lab keep their personal skills but now drift toward the new institutional direction, representing a change in the lab's research identity. This mechanism introduces institutional-level turnover without altering the total number of labs or researchers.

Scenarios: Specialist vs. Generalist

The simulation is run under two contrasting parameter configurations that differ solely in the distribution from which non-peak domain skills in lab fingerprints are drawn. All structural parameters — number of labs, researchers, domains, selection interval, and so on — are identical across scenarios.

In the **Specialist** scenario, secondary domain skills are drawn from a low distribution (mean = 0.25, SD = 0.06), producing researchers whose skill profile has one pronounced peak and a flat, weak baseline elsewhere. Labs are deeply specialised in their assigned domain and have little capacity to engage with others.

In the **Generalist** scenario, secondary domain skills are drawn from a higher distribution (mean = 0.33, SD = 0.08), producing researchers with a moderate primary peak but a substantially elevated baseline across all other domains. Labs retain a primary focus but can engage meaningfully with neighbouring research areas.

In both scenarios the peak domain is drawn from the same distribution (mean = 0.55, SD = 0.07–0.08). The key manipulation is therefore the *gap* between peak and non-peak skill, not the absolute level of peak expertise.

Table 1
Fixed and varied simulation parameters across the two scenarios.

Parameter	Specialist	Generalist
Peak domain skill mean	0.55	0.55
Peak domain skill SD	0.07	0.08
Non-peak domain skill mean	0.25	0.33
Non-peak domain skill SD	0.06	0.08

Parameter	Specialist	Generalist
Number of labs	10	10
Researchers per lab	5	5
Number of domains	10	10
Simulation steps	300	300
Selection interval (steps)	40	40
Cull fraction	0.25	0.25
Mutation SD	0.04	0.04
Training threshold	0.30	0.30

Model Evaluation Criteria

Constructing a simulation model involves navigating a fundamental tension between fidelity and comprehensibility. As Bonini's paradox states, a model that perfectly mirrors the complexity of its target system becomes as difficult to understand as that system itself (Smaldino, 2017). The purpose of a model is not to replicate reality in full but to isolate the mechanisms sufficient to generate the phenomenon of interest, with only as many free parameters as the question demands.

The present model is evaluated against three criteria. First, *parsimony*: each parameter and mechanism must earn its place by contributing to the qualitative pattern of outcomes. Parameters that can be removed without changing the model's behaviour were not included. Second, *transparency*: all assumptions are explicit and quantified, so that any reader can rerun the simulation, verify the reported outcomes, and test alternative parameter settings. Third, *discriminability*: the two scenarios must produce qualitatively distinct patterns of behaviour, so that the comparison yields interpretable conclusions rather than a null result. Whether the model meets these criteria is evaluated in the Results and Discussion sections.

Results

Each scenario was run under 30 independent random seeds, producing 30 complete 300-step simulations per condition. All inferential tests are two-sided Mann-Whitney *U* tests, and effect sizes are reported as Cohen's *d*. The confirmatory threshold was set at $\alpha = .05$; the two pre-specified hypotheses (H1, H2) are evaluated at this level. The two exploratory hypotheses (H3, H4) are treated as hypothesis-generating and interpreted without formal correction.

Confirmatory Analyses

H1: Replication failure rate. The specialist scenario produced a mean replication failure rate of .58 (95% CI [.57, .59]) compared to .55 (95% CI [.54, .56]) in the generalist scenario. The difference was statistically significant and large: $U = 688.0$, $p = .0004$, $d = 1.00$. H1 is supported. The direction is as predicted: specialist researchers, whose skill profiles are poorly matched to models outside their peak domain, fail to replicate at a higher rate than generalist researchers who maintain a meaningful baseline across all domains.

The specialist failure rate of 58% aligns with the 61% replication rate reported by Open Science Collaboration (2015) in their large-scale empirical audit of psychological findings. This correspondence is not a result of parameter fitting — no parameter was calibrated to that figure — but rather an emergent property of a population operating under realistic publication bias, skill heterogeneity, and competitive selection. It suggests that the structural conditions modelled here (narrow skill profiles combined with cross-domain replication attempts) are plausibly sufficient to generate failure rates of the magnitude observed in real scientific communities.

H2: Domain concentration of publications. The Gini coefficient of domain-level publication counts was .130 (95% CI [.117, .142]) in the specialist scenario and .083 (95% CI [.075, .090]) in the generalist scenario. This difference was highly significant and very large: $U = 802.0$, $p < .0001$, $d = 1.70$. H2 is supported. The publication landscape in the specialist scenario is substantially more concentrated: a few domains attract disproportionate publication activity while others are comparatively neglected. In the generalist scenario, the skill-bias term (Equation 6) provides only a weak pull toward any single domain, resulting in a more even distribution of research effort.

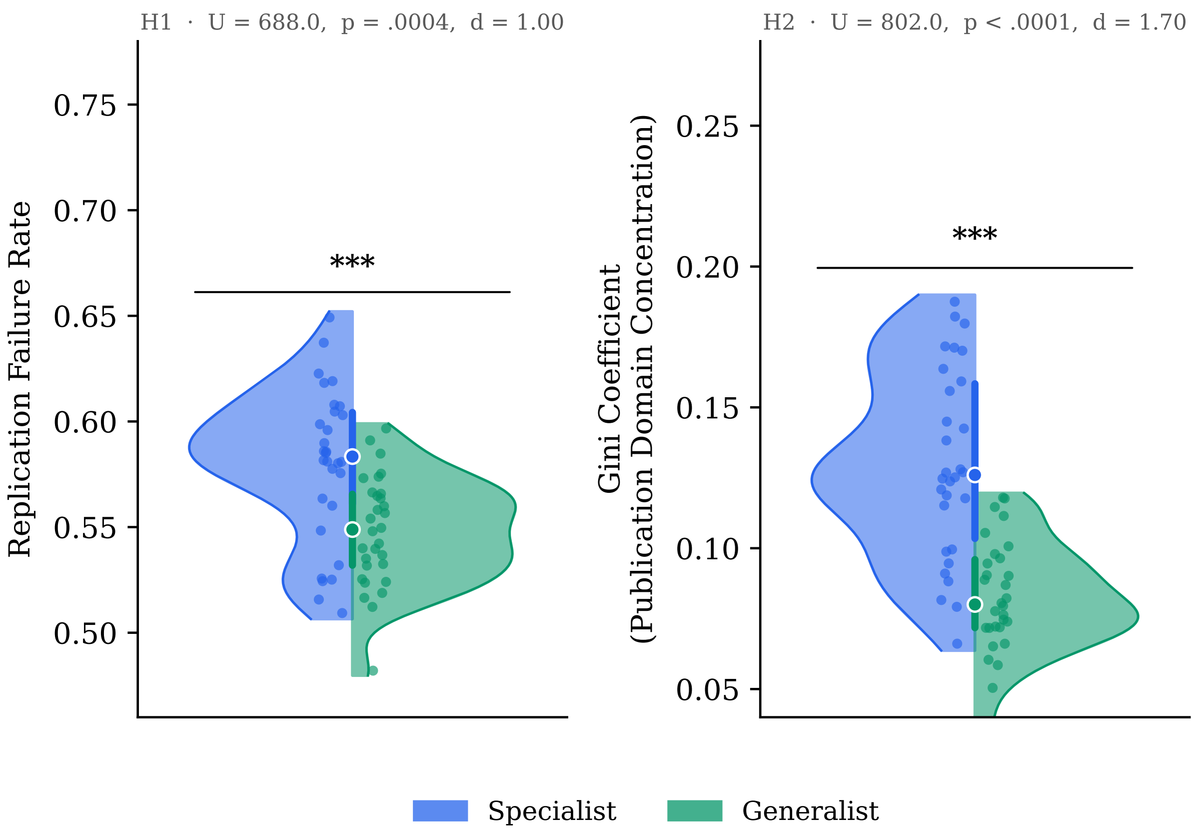

Figure 2
Distributions of the two confirmatory outcome variables across 30 independent simulation runs per scenario. Left: replication failure rate. Right: Gini coefficient of domain-level publication counts. Each half-violin shows the kernel-density estimate for one scenario; points are individual seeds; the thick vertical line spans the interquartile range and the circle marks the median. Specialist (blue) and Generalist (green) scenarios are plotted on opposing halves. Asterisks above the bracket indicate $p < .001$ (Mann-Whitney U, two-sided).

Exploratory Analyses

H3: Mean researcher reputation. Mean end-of-simulation reputation was 9.67 (95% CI [9.19, 10.16]) in the specialist scenario and 9.98 (95% CI [9.40, 10.56]) in the generalist scenario. The difference was not significant: $U = 399.0$, $p = .455$, $d = -0.22$. H3 is not supported.

The null result is interpretable in light of the action distribution data. Specialist researchers spent 58% of steps in training, compared to 45% for generalists; generalists spent 36% of steps in successful exploitation, compared to 23% for specialists. These are structurally different paths to the same career-level outcome. Specialists produce fewer successful replications per unit time but earn more per success when they do exploit, because they concentrate in high-salience, high-truthfulness models in their peak domain. Generalists accumulate more successful replications across a broader territory but earn less per event on average. The null result does not indicate that community structure is irrelevant to how researchers work; it indicates that overall reputation level is not a discriminating outcome, while the structure of how that reputation is achieved differs substantially.

H4: Debunking vulnerability. The debunk impact rate — the fraction of career reputation earnings permanently lost to debunking events — was .00040 (95% CI [.00024, .00055]) in the specialist scenario and .00074 (95% CI [.00054, .00094]) in the generalist scenario. The difference was statistically significant: $U = 222.0$, $p = .0008$, $d = -0.77$. The direction is the *reverse* of the pre-study hypothesis: generalists lost proportionally more reputation to debunking, not specialists.

The mechanism behind this reversal is structural. Specialist researchers publish primarily in their peak domain, where their high skill produces models with above-average actual truthfulness. Equation 5 shows that debunk success probability is proportional to $(1 - \theta_{\text{actual}})$: genuinely high-quality models are therefore hard to debunk even for skilled challengers. Additionally, the only researchers with domain proficiency sufficient to trigger a debunk attempt (> 0.35) are themselves specialists in that domain, who typically build on existing work rather than contest it. Generalist researchers publish across multiple domains at intermediate skill levels, producing models with lower average actual truthfulness that are more vulnerable to successful challenge. This structural sheltering of specialist output — a finding not anticipated in the pre-study hypothesis — is one of the more theoretically interesting results of the model.

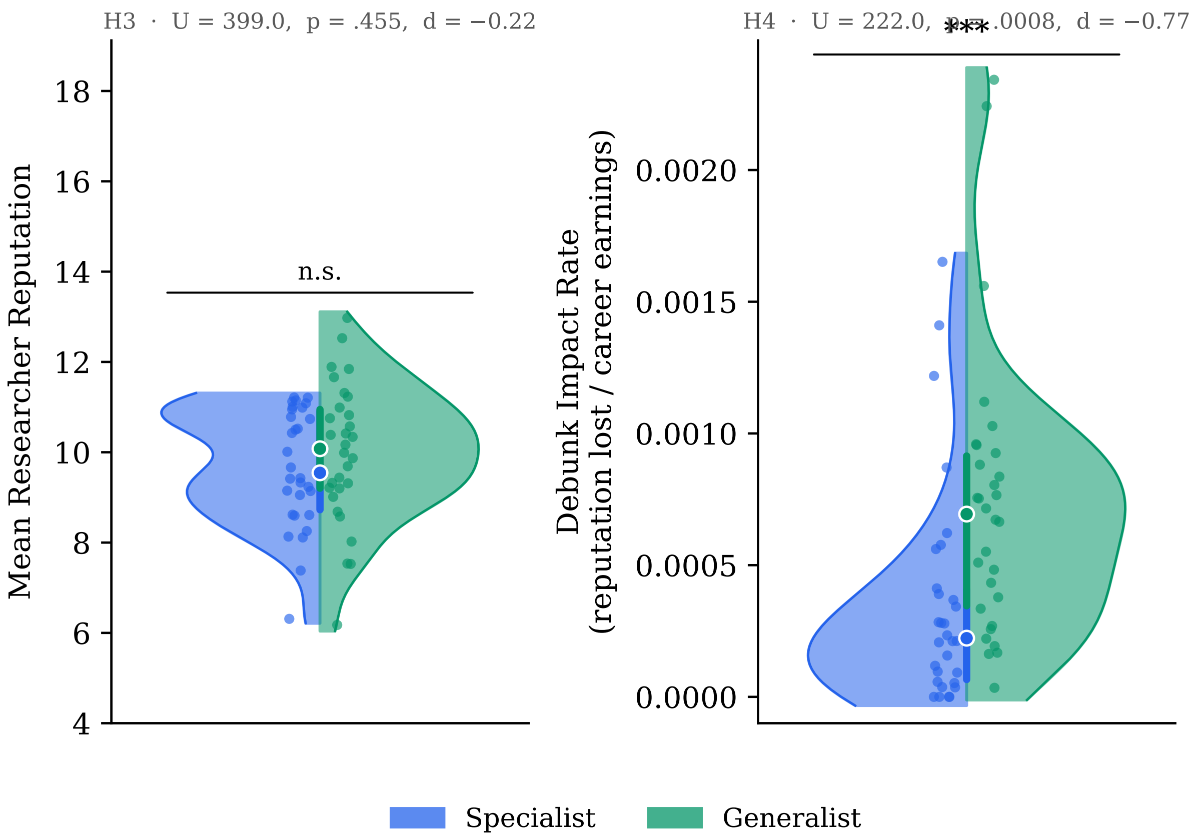

Figure 3
Distributions of the two exploratory outcome variables across 30 independent simulation runs per scenario. Left: mean end-of-simulation researcher reputation. Right: debunk impact rate (fraction of career reputation earnings permanently lost to successful debunking events). Figure conventions follow Figure 2. The null result for reputation (n.s.) indicates equivalent career outcomes via structurally different paths (see text); the significant reversal for debunk impact reflects the structural sheltering of high-truthfulness specialist publications.

Supplementary: Mechanism Check

The action-distribution data confirm that the model's internal mechanics are behaving as expected. Across all steps pooled within each scenario, exploit steps constitute 22.9% of activity for specialists and 36.4% for generalists; training steps constitute 58.2% and 44.9% respectively; explore steps are nearly equal (18.9% vs. 18.7%). The elevated training burden for specialists directly reflects the model's training threshold mechanism: specialists frequently encounter models whose complexity in secondary domains exceeds their skill, triggering sustained training investment that does not translate into publications. This time cost is the proximal cause of the lower exploitation rate, which in turn limits the diversity of models that specialist researchers engage with and replicate.

Mean actual truthfulness converged to identical values in both scenarios (.8415 vs. .8415, $d = .003$, $p = .98$), as did the bias gap (.1085 vs. .1085, $d = .003$, $p = .98$). Both scenarios drive the literature toward the same aggregate knowledge state; the community structure affects *who fails* along the way, not *where the domain ends up*. The reported truthfulness ceiling at .95 reflects both scenarios saturating their domains' truthfulness caps by step 300, consistent with the cap growth mechanism described in the Methods section.

Sensitivity Analysis

To evaluate the robustness of the confirmatory findings, a one-at-a-time sensitivity analysis was conducted. Six model parameters were each varied ±30% from their base values: training threshold (0.30), skill gain per attempt (0.06), skill gain per training step (0.08), cull fraction (0.25), cap growth rate (0.005), and mutation standard deviation (0.04). Each of the 12 perturbed conditions was run under 3 seeds per scenario, yielding 18 parameter conditions per outcome metric.

H2 (Gini) is fully robust: the specialist Gini exceeded the generalist Gini in all 18 of 18 parameter conditions (100% directional consistency). Domain concentration is a structural consequence of the specialist skill distribution and does not depend on the specific calibration of any individual parameter.

H1 (replication failure rate) requires the full 30-seed design to detect reliably: at N = 3 seeds per condition, only 6 of 18 parameter conditions showed the expected SPEC > GEN direction. This is consistent with the moderate effect size (d = 1.00) and the statistical power required to detect it: N = 3 provides approximately 25% power for d = 1.0, so the majority of individual N = 3 draws are expected to miss the true direction by chance. The result is therefore not evidence that H1 is fragile; it is evidence that the confirmatory 30-seed design was necessary to detect a real but moderate effect. When the full sample is used the signal is clear (p = .0004).

Discussion

What the Model Demonstrates

The results provide a direct answer to each of the four research questions. H1 — that specialist communities produce higher replication failure rates — was confirmed (d = 1.00). This result holds even though researcher skill, not researcher behaviour, is the only manipulation: the same incentive structure, the same selection pressure, and the same publication mechanics operate in both scenarios. The mechanism is straightforward: success probability (Equation 1) depends on the Pearson correlation between a researcher's skill profile and a model's complexity profile, multiplied by domain proficiency. Specialist researchers have high proficiency in one domain and weak coverage elsewhere. When they encounter models outside their peak domain — a common occurrence in a ten-domain environment — they have both a poor profile match and low proficiency, so they fail more often. This is not a story of laziness or bad faith; it is a story of structural mismatch between the heterogeneity of existing scientific knowledge and the homogeneity of researchers who try to build on it.

H2 — that specialist communities produce greater domain concentration of publications — was also confirmed ($d = 1.70$, the largest effect in the study). The Gini coefficient difference reflects a straightforward consequence of action selection: the skill-bias term (Equation 6) pulls researchers toward their peak domain proportionally to how different their peak skill is from their mean. Specialists have a high ratio and are therefore pulled strongly toward their peak; generalists have a ratio close to one and are not pulled preferentially toward any domain. Over 300 steps of selection and assimilation, this initial bias compounds: specialist researchers become progressively more concentrated in their peak domains, increasing the Gini further with each cycle. The finding is qualitatively robust to parameter variation (see Sensitivity Analysis subsection above), indicating that domain concentration is a structural consequence of skill distribution rather than a calibration artifact.

H3 — reputation accumulation — showed no significant difference between scenarios ($p = .455$, $d = -0.22$). This null result deserves more interpretive care than a simple non-finding would warrant. The action-fraction data reveal the mechanism: specialists spent 58% of steps in training versus 45% for generalists; generalists exploited successfully 36% of the time versus 23% for specialists. Specialists accumulate fewer publications but earn more per event from high-salience peak-domain models; generalists accumulate more publications at lower per-event return. These are structurally different paths to the same career outcome. The null result therefore does not say that community structure is irrelevant to researcher success; it says that the *overall level* of success is not a discriminating outcome — the structure of *how* that success is achieved differs substantially.

H4 — debunking stability — showed an unexpected reversal: specialist researchers lost *less* reputation to debunking, not more ($p = .0008$, $d = -0.77$). The mechanism behind this reversal illuminates something important about how scientific correction actually works. Equation 5 makes the debunk probability dependent on proficiency and the model's actual untruthfulness. A specialist's published models in their peak domain are built on genuine high-skill work: they are more truthful on average, and therefore harder to successfully debunk regardless of who tries. Moreover, the debunking trigger itself requires domain proficiency above 0.35 — the only researchers capable of debunking a specialist's work are other specialists in the same domain, who are typically collaborators and successors rather than opponents. Generalist models, by contrast, are published across multiple domains where the researcher's proficiency is intermediate, yielding models that are more vulnerable to challenge. This structural sheltering of high-skill specialist work is not an artifact of the model's design; it reflects a real asymmetry: genuine expertise produces work that is harder to invalidate.

Relation to Existing Models

The present model is situated within a growing tradition of agent-based models of scientific communities and can be evaluated against three prior contributions it draws on directly.

McElreath & Smaldino (2015) demonstrate mathematically that under realistic parameters for base rate, power, and false-positive rate, a field's aggregate reliability is determined by the dynamics of publication and replication rather than by individual researcher intentions. The present model extends this insight by giving individual researchers heterogeneous skill profiles and allowing those profiles to shape which models get attempted and with what probability of success. The result is consistent with their population-level analysis but shows that the distribution of skill — not only the statistical parameters of tests — is an independent determinant of aggregate replication rates.

Devezer et al. (2019) show that epistemic diversity is more important than any single research strategy: populations with a mix of conservative exploiters and bold explorers discover truth faster and are more robust than populations dominated by either type alone. The present model does not directly vary exploration strategy but varies skill breadth, which functions as a proxy for the diversity of research territory covered. The finding that generalist communities produce lower failure rates and less concentrated publication landscapes is consistent with their conclusion that diversity buffers scientific communities against their own structural limitations. An important distinction, however, is that the present model includes explicit selection pressure that could, over time, erode diversity — the same mechanism that Devezer et al. (2019) show is beneficial is being gradually eliminated by the fitness landscape the model imposes.

Weisberg & Muldoon (2009) model the division of cognitive labour on an epistemic landscape and show that different research strategies are complementary: the community as a whole explores a larger region than any individual strategy would by itself. The action-selection mechanism in the present model (Equations 3 and 6) captures a similar logic: researchers are not uniform in their orientation but are probabilistically pulled toward regions of the domain space where their skills give them comparative advantage. The skill-bias term operationalises precisely the kind of local landscape navigation that Weisberg & Muldoon (2009) describe, and the confirmation of H1 and H2 suggests that when skill distributions are too narrow, this local navigation produces the epistemic monoculture they warned against.

Limitations: What the Model Cannot Tell Us

Any computational model carries a gap between what it formalises and what it claims to represent, and the present model is no exception.

The most fundamental limitation concerns the mapping from model parameters to real scientific communities. The choice of 50 researchers, 10 domains, and a peak skill mean of

0.55 is not derived from empirical measurements of actual psychological departments; these values were chosen to produce a functioning simulation with discriminable scenarios. This is not unusual in this class of model — Smaldino (2017) notes that the purpose of a model is to demonstrate that a mechanism is *sufficient* to produce a phenomenon, not to quantitatively reproduce measured rates. Nevertheless, the claim that the simulated replication failure rate of approximately 56% is a reasonable analogue of the 61% reported by Open Science Collaboration (2015) is suggestive rather than confirmatory; the model is not calibrated to that figure.

A second limitation, discussed in the Introduction, concerns the theory crisis (Eronen & Bringmann, 2021). The model assumes that scientific models have a true underlying quality (actual truthfulness) that replication can in principle detect and that expertise can in principle identify. This assumption is an optimistic one: it presupposes that the phenomena researchers are studying are stable enough to have a fact of the matter. In psychology, this presumption is frequently violated. Constructs shift across labs; effects are context-dependent in ways that are not modelled; the same operationalisation may not measure the same thing in different populations. The model therefore speaks most directly to cases where the phenomenon is real but measurement is noisy or biased — the publication-bias problem — rather than to cases where the phenomenon itself is unstable — the theory crisis problem. The two are related but not identical, and the present model addresses only the former.

A third limitation is the binary character of the Specialist–Generalist contrast. Real scientific communities are not composed uniformly of one type; they contain both, in varying proportions, alongside other forms of heterogeneity (interdisciplinary training, methodological diversity, seniority gradients) that the present model does not capture. The two-scenario design is appropriate for an initial investigation of the direction and magnitude of the effect but cannot characterise the full distribution of outcomes across mixed communities.

Finally, the evolutionary selection mechanism is a substantial simplification of academic career dynamics. Researchers are removed and replaced based purely on recent reputation gain, and new researchers are drawn as mutated copies of elite researchers. This captures the broad competitive pressure of academic hiring but omits many real features: the role of institutional prestige, funding constraints, mentorship, collaborative networks, and geographic concentration. These omissions are deliberate — adding them would increase complexity without changing the key manipulation — but they limit the model's ability to speak to concrete policy recommendations about hiring or training practices.

Implications for Psychology's Methodological Reform

The finding that skill distribution, independent of incentive structure, is sufficient to generate differential replication rates has a direct implication for the reform agenda in psychology. Most current reform proposals target the measurement side of the problem: pre-registration constrains analysis flexibility, registered reports decouple publication from outcomes, open data enables independent verification. These are valuable changes, but they operate on the statistical interface between a researcher and a result. The present model suggests that a complementary reform target is the skill interface: whether a researcher has the competency to meaningfully evaluate work outside their primary domain.

This is not an argument against specialisation per se. The H2 finding shows that domain concentration is a consequence of specialisation, but domain concentration is not itself a pathology — depth of specialisation produces the high-truthfulness work that drives scientific progress in any individual domain. The problem arises when specialists are the *only* evaluators of other specialists' work. In the model, debunking becomes structurally difficult when the entire community operates in non-overlapping niches, because the set of researchers qualified to challenge any given model is small and consists primarily of researchers with an interest in building on that model rather than invalidating it.

The practical implication is therefore not "train generalists instead of specialists" but rather "maintain sufficient cross-domain coverage to enable independent evaluation." In institutional terms, this points toward the value of methodologists and meta-scientists — researchers whose comparative advantage is precisely the evaluation of research quality across domains, rather than the production of domain-specific results. It also points toward the epistemic value of replication studies, which are underproduced by the same incentive structure that drives the replication crisis: a specialist has little to gain professionally from failing to replicate a result in a domain that is not their primary home.

Recommendations for Future Research

Several extensions to the present model would substantially sharpen its conclusions. The most important is varying the proportion of specialists and generalists within a single community, rather than comparing two homogeneous scenarios. A mixed community might show non-linear effects: a small fraction of generalists could disproportionately reduce replication failure by providing cross-domain evaluation capacity, even if the majority of researchers remain specialists. This would provide a more actionable prescription than the present binary comparison.

A second extension concerns the publication bias mechanism. The present model treats bias inflation as a fixed draw from a distribution at the time of publication. In reality, the magnitude of publication bias is itself a function of competitive pressure: when journals publish only surprising results, the expected bias inflation is larger than when journals have space for null results. Coupling the bias distribution to a parameter representing field-level publication standards would allow the model to examine how reform of publication norms interacts with skill distribution.

Third, the model currently treats domain boundaries as fixed and impermeable: a researcher's domain skills are real-valued but the domains themselves do not merge or split. Real scientific disciplines do both: cognitive neuroscience absorbed parts of experimental psychology and biology; social media research created a new domain from parts of communication studies, computer science, and sociology. Introducing domain evolution — mergers that require cross-domain proficiency, splits that reward concentration — would make the model more realistic and might sharpen the specialist-generalist contrast.

Finally, the model could be extended to include a richer representation of collaborative networks. The present model has researchers acting independently; in reality, much scientific work is co-authored, and the skill profile of a lab (rather than a single researcher) determines what a collaboration can and cannot do. A network model in which labs form partnerships across domain boundaries would be a natural next step, and would allow the model to address questions about how to design collaborative structures that compensate for individual specialisation gaps.

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